

Statistics: Preliminary Ph.D. Course

Day 4: Parametric Distributions and the Conditional Expectation

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Today's Session

We return to Hansen's Book

- **Parametric Distributions** and the **Parameter Space**. Definitions.
- Some **Important Distributions**. Bernoulli, Binomial, Uniform, Normal, Logistic, Chi-Square, Cauchy, and Student-t distributions. PMFs, PDFs, Moments, and Applications. Important Results.
- **Conditional Distributions**. Continuous and Discrete cases. Conditioning and Independence. Examples.
- **Bayes' Theorem** for Densities. Examples.
- The **Conditional Expectation**. The **Law of Iterated Expectations (LIE)**. Existence and Uniqueness. **Identification**. The CEF and the **MSE**. Examples.

Parametric Distributions and the Parameter Space

Definitions

- **Definition 4.1:** A **Parametric Distribution** $F(x; \theta)$ is a distribution indexed by a **Parameter** $\theta \in \Theta$.
- **Definition 4.2:** The set Θ is called the **Parameter Space**, i.e., the set of possible values of the parameter θ .
- For each θ , $F(x; \theta)$ is a valid distribution function. As θ varies, $F(x; \theta)$ changes.
- We sometimes call $F(x; \theta)$ a **Family of Distributions**.
- Econometricians use parametric distributions for statistical modeling. Typically, a set of observations is modeled using a specific distribution with parameters estimated from the data.

Parametric Distributions and the Parameter Space

Bernoulli Distribution

- **Definition 4.3:** A **Bernoulli** random variable X is a discrete, two-point distribution $\mathcal{X} = \{0, 1\}$ with one parameter, p . We write $X \sim \text{Ber}(p)$:

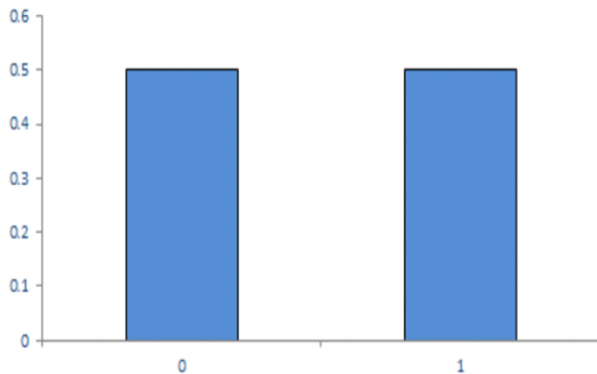
$$\mathbb{P}(X = 1) = p \text{ and } \mathbb{P}(X = 0) = 1 - p$$

- Its **PMF** is given by: $\pi(x; p) = \mathbb{P}(X = x; p) = p^x(1 - p)^{1-x}$. Why?
- The **parameter space** is $p \in (0, 1)$. It is appropriate for any random variable which has only two outcomes.
- $\mathbb{E}[X] = p$ and $\mathbb{V}ar[X] = p(1 - p)$. **Proof.**

Example. (Potential Outcomes Model and the Treatment Variable).

Parametric Distributions and the Parameter Space

Bernoulli Distribution: An Illustration with $p = 0.5$



Parametric Distributions and the Parameter Space

Binomial Distribution

- **Definition 4.4:** A **Binomial** random variable X has a discrete distribution with support $\mathcal{X} = \{0\} \cup \mathbb{N}$, parameters $(n, p)'$, and **PMF**:

$$\pi(x; n, p) = \binom{n}{x} p^x (1 - p)^{n-x}$$

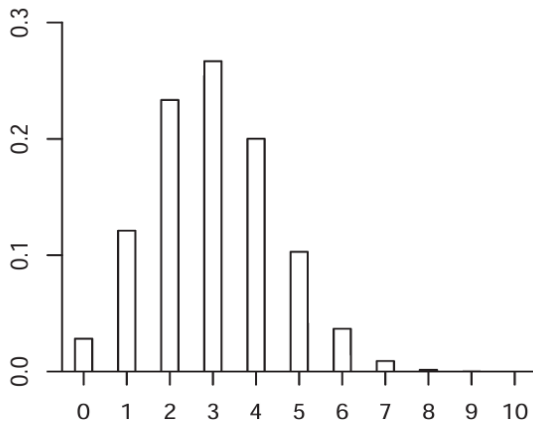
We write $X \sim Bi(n, p)$.

- The **parameter space** is $(n, p)' \in \mathbb{N} \times (0, 1)$. It is appropriate for the sum of n independent Bernoulli trials.
- $\mathbb{E}[X] = np$ and $\mathbb{V}ar[X] = np(1 - p)$. **Proof.**
- **CDF:** $F(x; p) = \sum_{i=0}^{\lfloor x \rfloor} \binom{n}{i} p^i (1 - p)^{n-i}$

Example. (Potential Outcomes Model and the Treatment Variable).

Parametric Distributions and the Parameter Space

Binomial Distribution: An Illustration with $p = 0.3$ and $n = 10$



Parametric Distributions and the Parameter Space

Uniform Distribution (Continuous)

- **Definition 4.5:** A **Uniform** random variable X has a continuous distribution with two parameters $(a, b)'$ (minimum and maximum), support $\mathcal{X} = [a, b] \in \mathbb{R}$, and **PDF**:

$$f(x; a, b) = \frac{1}{b - a} \mathbb{1}_{[a, b]}(x)$$

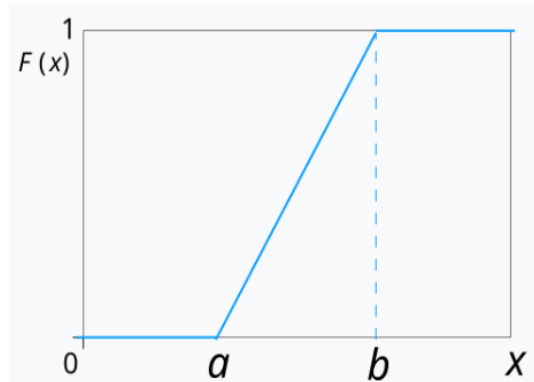
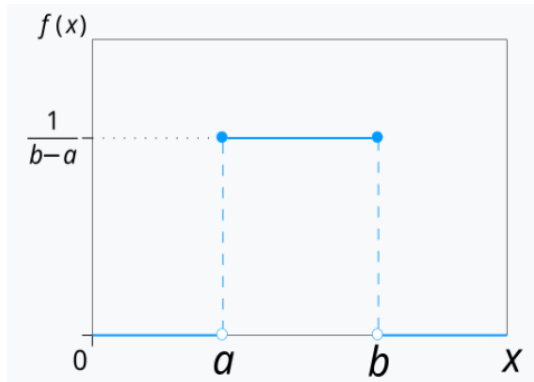
We write $X \sim U[a, b]$.

- The **parameter space** is $(a, b)' \in \{(a, b) \in \mathbb{R}^2 : a < b\}$.
- $\mathbb{E}[X] = (a + b)/2$ and $\text{Var}[X] = (b - a)^2/12$. **Proof.**
- **CDF:** $F(x; a, b) = \frac{x-a}{b-a} \mathbb{1}_{[a, b]}(x)$.

Example. (Distribution of a PIT and random number generation in a computer).

Parametric Distributions and the Parameter Space

Uniform Distribution: An Illustration



Parametric Distributions and the Parameter Space

Normal Distribution

- **Definition 4.6:** A **Normal** random variable X has a continuous distribution with two parameters $(\mu, \sigma^2)'$ (location and scale), support $\mathcal{X} = \mathbb{R}$, and **PDF**:

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(\frac{-(x - \mu)^2}{2\sigma^2}\right), \text{ denoted by } \phi(x) \text{ when } (\mu, \sigma^2)' = (0, 1)'.$$

We write $X \sim N(\mu, \sigma^2)$.

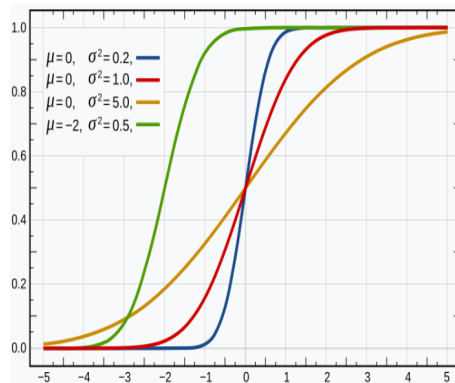
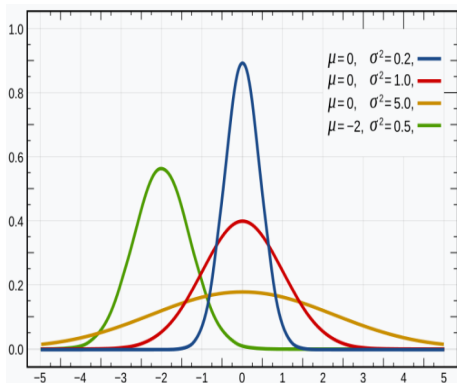
- The **parameter space** is $(\mu, \sigma^2)' \in \mathbb{R} \times \mathbb{R}_{++}$.
- $\mathbb{E}[X] = \mu$ and $\mathbb{V}ar[X] = \sigma^2$. **Proof.** (Hint: The MGF is $M(t) = \exp(\mu t + \sigma^2 t^2/2)$).
- **CDF:**

$$F(x; \mu, \sigma^2) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}\sigma} \exp\left(\frac{-(x - \mu)^2}{2\sigma^2}\right) dx, \text{ denoted by } \Phi(x) \text{ when } (\mu, \sigma^2)' = (0, 1)'.$$

Example. Critical values in hypothesis testing. Probit Models.

Parametric Distributions and the Parameter Space

Normal Distribution: An Illustration



Parametric Distributions and the Parameter Space

Multivariate Normal Distribution

- Let $Z = (Z_1, \dots, Z_m)'$ be a random vector with mutually independent elements, where $Z_j \sim N(0, 1) \quad \forall j = 1, \dots, m$. The vector Z has a **Multivariate Standard Normal Distribution**, written $Z \sim N(0, I_m)$, with density:

$$\phi(Z) = \frac{1}{(2\pi)^{m/2}} \exp\left(-\frac{z'z}{2}\right)$$

Proof. (Hint: Use independence and collect terms, then use vector notation).

- If $Z \sim N(0, I_m)$ and $X = \mu + BZ$, where B is a $q \times m$ matrix and μ is a $q \times 1$ vector, then X has the **Multivariate Normal Distribution**, written $X \sim N(\mu, \Sigma)$, with $q \times q$ variance-covariance matrix $\Sigma = BB'$. If Σ is invertible, the density of X is:

$$\phi(x) = \frac{1}{(2\pi)^{m/2} \det(\Sigma)^{1/2}} \exp\left(-\frac{(x - \mu)' \Sigma^{-1} (x - \mu)}{2}\right)$$

Proof. (Hint: Use the formula for monotonic transformations in Theorem 2.22).

Parametric Distributions and the Parameter Space

Standard Logistic Distribution

- **Definition 4.7:** A **Logistic** random variable has a continuous distribution with two parameters $(\mu, s)' \in \mathbb{R} \times \mathbb{R}_{++}$ (location and scale) and support $\mathcal{X} = \mathbb{R}$. In econometrics, the standard logistic $(\mu, s)' = (0, 1)'$ is of particular interest: it has the following convenient **CDF**:

$$F(x) = \Lambda(x) = \frac{1}{1 + e^{-x}}$$

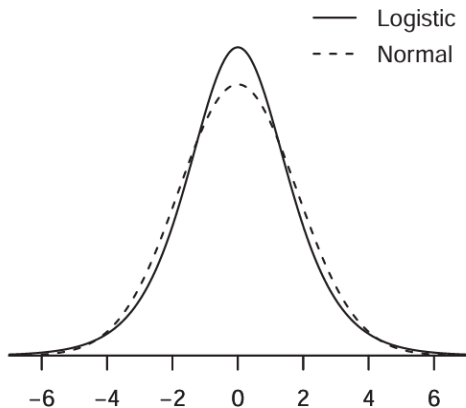
We write $X \sim \Lambda$.

- $\mathbb{E}[X] = 0$ and $\mathbb{V}ar[X] = \pi^2/3$. **Proof.** (Hint: You can find the proof [here](#). The variance is hard to show).
- **PDF:** $f(x) = F(x)(1 - F(x))$.

Example. (Logistic Regression).

Parametric Distributions and the Parameter Space

Standard Logistic Distribution: An Illustration



Parametric Distributions and the Parameter Space

Chi-Square Distribution

- **Definition 4.8:** A **Chi-Square** random variable X , has a continuous distribution with one parameter ν (degrees of freedom), parameter space $\nu \in \mathbb{N}$, and **PDF**:

$$f(x; \nu) = \frac{1}{2^{\nu/2} \Gamma(\nu/2)} x^{\nu/2-1} \exp(-x/2)$$

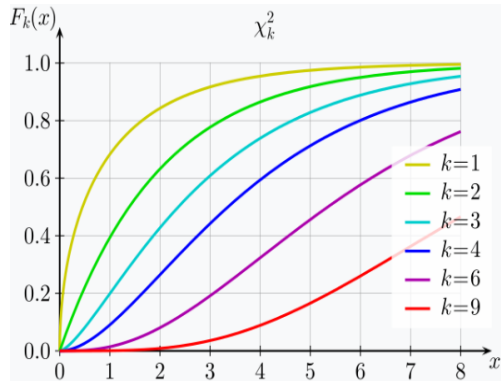
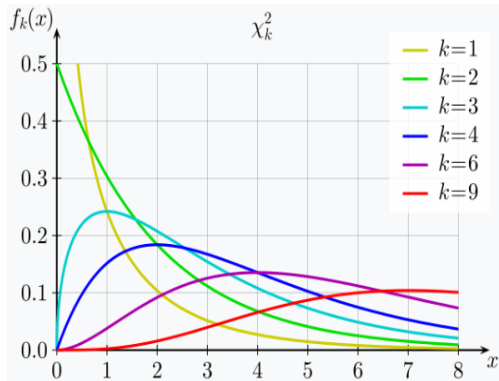
where $\Gamma(\cdot)$ is the Gamma Function. We write $X \sim \chi^2_{(\nu)}$.

- It has **support** $\mathcal{X} = \mathbb{R}_+$ if $\nu > 1$ and $\mathcal{X} = \mathbb{R}_{++}$ if $\nu = 1$.
- $\mathbb{E}[X] = \nu$ and $\mathbb{V}ar[X] = 2\nu$. **Proof.** (Hint: The ChF is $ChF(t) = (1 - 2it)^{-\nu/2}$).
- **CDF:** complicated.

Example. Critical values in hypothesis testing.

Parametric Distributions and the Parameter Space

Normal Distribution: An Illustration



Parametric Distributions and the Parameter Space

Cauchy Distribution

- **Definition 4.9:** A **Cauchy** random variable X , has a continuous distribution with two parameters $(\lambda, \gamma)'$ (location and scale), support $\mathcal{X} = \mathbb{R}$, and **PDF**:

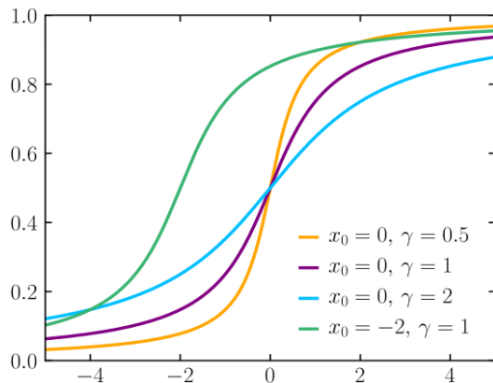
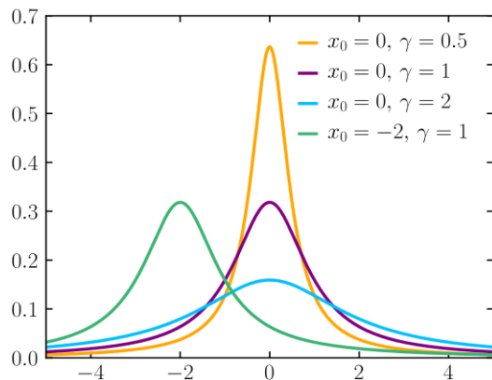
$$f(x; \lambda, \gamma) = \frac{1}{\pi\gamma \left[1 + \left(\frac{x-\lambda}{\gamma} \right)^2 \right]}$$

We write $X \sim \text{Cauchy}(\lambda, \gamma)$.

- The **parameter space** is $(\lambda, \gamma)' \in \mathbb{R} \times \mathbb{R}_{++}$.
- The distribution has very **fat tails** and no moments.
- $\mathbb{E}[X] = \infty - \infty$. **Proof.**
- **CDF:** $F(x; \lambda, \gamma) = \frac{1}{\pi} \arctan \left(\frac{x-\lambda}{\gamma} \right) + \frac{1}{2}$.

Parametric Distributions and the Parameter Space

Cauchy Distribution: An Illustration



Parametric Distributions and the Parameter Space

Student-t Distribution

- **Definition 4.10:** A **Student-t** random variable X , has a continuous distribution with one parameter ν (degrees of freedom), support $\mathcal{X} = \mathbb{R}$, and **PDF**:

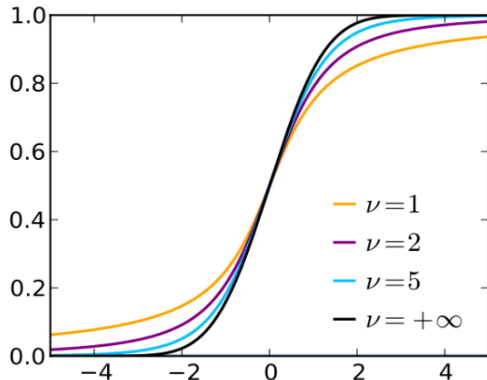
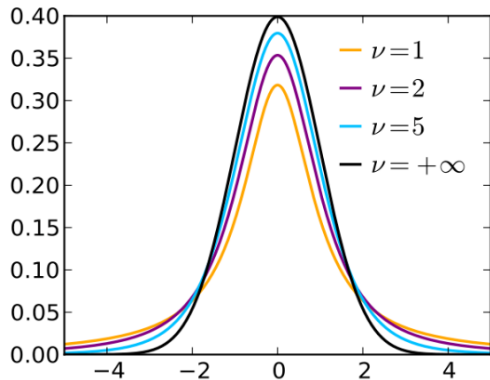
$$f(x; \nu) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\Gamma(\nu/2)} \left(1 + \frac{x^2}{\nu}\right)^{-\frac{(\nu+1)}{2}}$$

where $\Gamma(\cdot)$ is the Gamma Function. We write $X \sim t_\nu$.

- The **parameter space** is $(\nu)' \in \mathbb{R}_{++}$.
- $\mathbb{E}[X] = 0$ for $\nu > 1$, otherwise undefined. $\mathbb{V}ar[X] = \frac{\nu}{\nu-2}$ for $\nu > 2$, ∞ for $\nu \in (1, 2]$, otherwise undefined.
- **CDF**: complicated.

Parametric Distributions and the Parameter Space

Student-t Distribution: An Illustration



Parametric Distributions and the Parameter Space

Important Results

Theorem 4.1 (Student-t with Infinite DoF is Standard Normal)

As $\nu \rightarrow \infty$ the Student-t distribution becomes a Standard Normal distribution.

Proof. (See pg.70 in Hansen's Book).

Proposition 4.1 (Student-t with One DoF is Standard Cauchy)

When $\nu = 1$ the Student-t distribution specializes to a Standard Cauchy distribution.

Proof. (Hint: Use the density of the Student-t and the fact that $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ and $\Gamma(1) = 1$).

Theorem 4.2 (Uncorrelation and Normality)

If (X, Y) are Multivariate Normal with $\text{Cov}(X, Y) = 0$ then X and Y are independent.

Proof. (Hint: Use independence and the fact that uncorrelation implies that Σ is block diagonal).

Parametric Distributions and the Parameter Space

Important Results

Theorem 4.3 (A Linear Combination of Normals is also Normal)

If $X \sim N(\mu, \Sigma)$ then $Y = a + BX \sim N(a + B\mu, B\Sigma B')$

Proof. (Hint: Use the multivariate MGF of Y).

Theorem 4.4 (Standard Normals and Chi-Square I)

Let $Z \sim N(0, I_m)$. Then $Z'Z \sim \chi_m^2$

Proof. (See pg. 123 in Hansen's Book).

Theorem 4.5 (Standard Normals and Chi-Square II)

Let $X \sim N(0, A)$, where A is an $m \times m$ non-singular matrix. Then, $X'A^{-1}X \sim \chi_m^2$

Proof. (See pg. 123 in Hansen's Book).

Parametric Distributions and the Parameter Space

Important Results

Theorem 4.6 (Standard Normal, Chi-Square and Student-t)

Let $Z \sim N(0, 1)$ and let $Q \sim \chi^2_\nu$. Then

$$T = \frac{Z}{\sqrt{\frac{Q}{\nu}}} \sim t_\nu$$

Proof. (See pg. 123 in Hansen's Book).

Theorem 4.7 (Standard Normals and Cauchy)

Let $C = \frac{Z_1}{Z_2}$, where $Z_j \sim N(0, 1)$ for $j \in \{1, 2\}$. Then $C \sim \text{Cauchy}(0, 1)$.

Proof. (Hint: Use Theorem 4.6 and the fact that t_1 is a Standard Cauchy).

Conditional Distributions

Discrete and Continuous Cases: Definitions

- **Definition 4.11:** If X has a discrete distribution, the **Conditional Distribution Function** of Y given $X = x$ is $F_{Y|X=x}(y) = \mathbb{P}(Y \leq y | X = x)$ for any x such that $\mathbb{P}(X = x) > 0$.
- You can think of it as the distribution function for the **subpopulation** where $X = x$.
Example.
- **Definition 4.12:** If $F_{Y|X=x}(y)$ is differentiable w.r.t. y and $\mathbb{P}(X = x) > 0$ then the **Conditional Density Function** of Y given $X = x$ is $f_{Y|X=x}(y) = \frac{\partial}{\partial y} F_{Y|X=x}(y)$.
- **Definition 4.13:** For continuous X and Y the **Conditional Distribution Function** of Y given $X = x$ is

$$F_{Y|X=x}(y) = \lim_{\epsilon \rightarrow 0^+} \mathbb{P}(Y \leq y | x - \epsilon \leq X \leq x + \epsilon)$$

Fortunately, the expression can be simplified.

Conditional Distributions

Definitions

Theorem 4.8 (Continuous Conditional Distribution)

If $F(x, y)$ is differentiable with respect to x and $f_X(x) > 0$ then

$$F_{Y|X=x}(y) = \frac{\frac{\partial}{\partial x} F(x, y)}{f_X(x)}$$

Proof. (Hint: Use the definition of conditional distribution, Bayes' Theorem, and the definition of derivative).

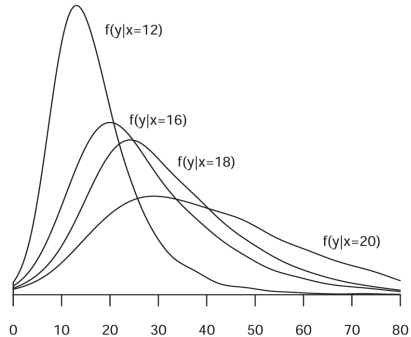
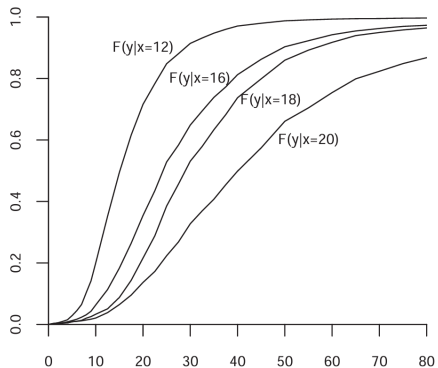
- **Definition 4.14:** For continuous X and Y the **Conditional Density Function** of Y given X is

$$f_{Y|X=x}(y) = \frac{\partial}{\partial y} F_{Y|X=x}(y) = \frac{\partial}{\partial y} \frac{\frac{\partial}{\partial x} F(x, y)}{f_X(x)} = \frac{f(x, y)}{f_X(x)}$$

for any x such that $f_X(x) > 0$.

Conditional Distributions

An Illustration



Conditional Distributions

Definitions

Theorem 4.9 (Conditional Densities and Independence)

If X and Y are independent and continuously distributed, $f_{Y|X}(y) = f_Y(y)$ and $f_{X|Y}(x) = f_X(x)$.

Proof. (Hint: Use the definition of conditional density and independence).

Theorem 4.10 (Bayes Theorem for Densities)

$$f_{Y|X}(y) = \frac{f_{X|Y}(x)f_Y(y)}{f_X(x)} = \frac{f_{X|Y}(x)f_Y(y)}{\int_{\mathbb{R}} f_{X|Y=y}(x)f_Y(y)dy}$$

Proof. (Hint: Use the definitions of conditional density and marginal distribution).

Conditional Expectation

Definition

- It is a **key** concept in econometrics.
- **Definition 4.15:** The **Conditional Expectation** of Y given $X = x$ is the expected value of the conditional distribution $F_{Y|X}(y)$ and is written as $m(x) = \mathbb{E}[Y|X = x]$. For discrete random variables this is:

$$\mathbb{E}[Y|X = x] = \sum_j \tau_j \frac{\pi(x, \tau_j)}{\pi_X(x)}$$

and for continuous Y this is:

$$\mathbb{E}[Y|X = x] = \int_{\mathbb{R}} y f_{Y|X=x}(y) dy$$

We also call $\mathbb{E}[Y|X = x]$ the conditional mean.

Examples.

Conditional Expectation

Properties

Lemma 4.1

Let X be integrable and let g be some integrable function. Then: $\mathbb{E}[g(X)|X] = g(X)$.

Lemma 4.2

Let Y be integrable. Then: $\mathbb{E}[g(X) + Y|X] = g(X) + E[Y|X]$ and $\mathbb{E}[g(X)Y|X] = g(X)E[Y|X]$.

Lemma 4.3

The conditional expectation $E[Y|X]$ satisfies Jensen's Inequality.

Examples.

Conditional Expectation

The Law of Iterated Expectations (LIE)

- Notice that $m(x)$ is not random, but $m(X)$ is. The Conditional Expectation $\mathbb{E}[Y|X]$ is a random variable. The following is a critical result that you will be using *ad nauseam*.

Theorem 4.11 (Law of Iterated Expectations, LIE)

If $\mathbb{E}[|Y|] < \infty$ then $\mathbb{E}[\mathbb{E}[Y|X]] = \mathbb{E}[Y]$.

Proof. (Hint: Use the definitions of expectation, conditional expectation, and conditional density/probability mass function).

Examples.

Property 4.1 (The Towering Property of LIE)

If $\mathbb{E}[|Y|] < \infty$ then $\mathbb{E}[Y|X] = \mathbb{E}[\mathbb{E}[Y|X, Z] | X]$.

Examples.

- In words, the average across group averages is the grand average.

Conditional Expectation

Existence and Uniqueness

Theorem 4.12 (Existence and Uniqueness of the Conditional Expectation)

Let (X,Y) have a joint distribution. If $\mathbb{E}[|Y|] < \infty$ then there exists a function $m(x)$ such that for all sets A for which $\mathbb{P}(X \in A)$ is defined:

$$\mathbb{E}[\mathbb{1}_{\{X \in A\}} Y] = \mathbb{E}[\mathbb{1}_{\{X \in A\}} m(X)]$$

Moreover, the function $m(X)$ is almost everywhere unique, in the sense that if $h(x)$ satisfies the above, then there is a set S such that $\mathbb{P}(S) = 1$, and $h(x) = m(x)$ for $x \in S$. The functions $m(x) = \mathbb{E}[Y|X = x]$ and $m(X) = \mathbb{E}[Y|X]$ are called the Conditional Expectation.

We will not prove this theorem, but the proof uses the Radon-Nikodým Theorem.

Conditional Expectation

Identification

- A critical issue in structural econometric modeling is **identification**, i.e., that a parameter of interest is uniquely determined by the distribution of observables.
- Let F denote a probability distribution, e.g., the distribution of (X, Y) , and let $\mathcal{F}$ be a collection of distributions. Let θ be the parameter of interest, e.g., $\mathbb{E}[Y]$.
- **Definition 4.16:** A parameter $\theta \in \mathbb{R}^k$ is **Identified** on $\mathcal{F}$ if for all $F \in \mathcal{F}$ there is a unique value of θ .

Equivalently, θ is identified if we can write it as a mapping $\theta = g(F)$ on the set $\mathcal{F}$.

Examples.

Theorem 4.13 (Identification of the Conditional Expectation)

Let (X, Y) have a joint distribution. If $\mathbb{E}[|Y|] < \infty$, the conditional expectation $m(X) = \mathbb{E}[Y|X]$ is identified almost everywhere.

Examples.

Conditional Expectation

Best MSE-Predictor

- Let X and Y be two random variables, where Y is square-integrable, i.e., $\mathbb{E}[Y^2] < \infty$.
- One may wish to predict the value of Y based on observed values of X . Let $g(X)$ be the predictor. Note that $g \in \mathfrak{N} = \{\text{all Borel functions } g \text{ with } \mathbb{E}[g(X)^2] < \infty\}$.
- Suppose that each predictor is assessed by the **Mean Squared Error (MSE)**, ie., by $\mathbb{E}[(Y - g(X))^2]$.

Theorem 4.14 (The CEF minimizes the MSE)

The Conditional Expectation Function (CEF) $g(X) = \mathbb{E}[Y|X]$ is the best predictor of Y in the sense that

$$\mathbb{E}[(Y - \mathbb{E}[Y|X])^2] = \min_{g \in \mathfrak{N}} \mathbb{E}[(Y - g(X))^2]$$

Proof. (Hint: Add and subtract the CEF, then use the binomial expansion and use both the LIE and the properties of the CEF). See Shao Example 1.19.

Recommended Exercises

Hansen's Book

- Chapter 3, Exercises: 3.1-3.5, 3.7b, 3.7c, and 3.12.
- Chapter 4, Exercises: 4.1d, 4.2d-e, 4.3d-e, 4.13c, 4.15, and 4.16.
- Chapter 5, Exercises: 5.1-5.7, 5.10-5.15, 5.17-5.19.

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